



systemgmmkit: Reproducible Dynamic-Panel GMM Workflows in Python

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Abstract

systemgmmkit is a Python toolkit for static panel estimators, instrumental-variable estimation, Difference GMM, System GMM, diagnostics, post-estimation, panel-aware validation, forecasting, and diagnostic-constrained specification search. The package does not introduce a new GMM estimator. Its contribution is an integrated and auditable workflow in which model specification, instrument construction, finite-sample covariance choices, diagnostics, predictions, and reporting remain visible. This paper documents release 1.0.0, including structured instrument-health reporting and the six-specification numerical certificate established in release 0.5.14 against Stata `xtabond2`, nested leakage-controlled N-CMAPSS results, an executed automatic GMM search, and Universal Output Hub table generation. The reported parity is benchmark-specific and must not be read as evidence that every empirical specification is valid.

Keywords: dynamic panel data, generalized method of moments, System GMM, Python, reproducibility.

1. Introduction

Dynamic-panel models are used when outcomes are persistent, unit-specific effects matter, and one or more regressors may be predetermined or endogenous (Arellano and Bond 1991; Blundell and Bond 1998; Bond 2002). In practice, applied work requires more than a coefficient vector. Researchers must control the estimation sample, define instrument roles and lag windows, monitor instrument growth, assess serial-correlation and over-identification diagnostics, and preserve enough metadata for independent reproduction.

systemgmmkit brings these tasks into a common Python interface. The package covers OLS, pooled OLS, fixed effects, random effects, panel IV/2SLS, Difference GMM, and System GMM, with structured result objects and post-estimation utilities. It complements estab-

lished software rather than ranking or replacing it (Roodman 2009; Croissant and Millo 2008; Seabold and Perktold 2010; Sheppard 2024; pydynpd developers 2024).

2. Software design and claim boundaries

The public workflow separates model specification, estimation, diagnostics, prediction, and reporting. For dynamic GMM, users declare the dependent variable, panel indexes, regressor roles, lag windows, transformation, collapsed or uncollapsed instruments, estimation step, and covariance correction. Results retain observation and group counts, instrument counts, coefficient and covariance output, Hansen and Sargan tests, and Arellano–Bond serial-correlation diagnostics.

Version 1.0.0 declares the documented public interface stable and adds a structured instrument-health assessment to native and `pydynpd` result summaries. It reports instrument and group counts, their ratio, and conservative acceptable, approaching, critical, or unavailable states. A critical state means that instruments outnumber groups and prompts shorter lag windows, collapsed instruments, and sensitivity checks. This rule is a screening signal rather than proof of bias or automatic invalidation of Hansen/Sargan inference.

Release 0.5.14 added certified unbalanced-panel and variable-specific missing-data paths to the maintained `xtabond2` evidence. The release certificate tests complete parameter sets, Windmeijer-corrected two-step standard errors, counts, diagnostics, and, where relevant, exact estimation-sample keys. Agreement is therefore asserted only for the registered benchmark specifications. A passing parity row says that two implementations agree; it does not say that the instruments are substantively valid or that the specification answers a particular research question.

3. Econometric formulation

The package represents the dynamic panel model as

$$y_{it} = \rho y_{i,t-1} + \mathbf{x}_{it}^\top \boldsymbol{\beta} + \alpha_i + \tau_t + \varepsilon_{it}, \quad (1)$$

where $i = 1, \dots, N$ indexes units, $t = 1, \dots, T_i$ indexes time, α_i is an unobserved unit effect, and τ_t is an optional time effect. The regressor vector $\mathbf{x}_{it}$ is partitioned by the user’s endogenous, predetermined, and exogenous declarations. Setting $\rho = 0$ yields the static-panel core; the role declarations and lag windows determine the dynamic-GMM instrument design.

First differencing removes the time-invariant unit effect:

$$\Delta y_{it} = \rho \Delta y_{i,t-1} + \Delta \mathbf{x}_{it}^\top \boldsymbol{\beta} + \Delta \tau_t + \Delta \varepsilon_{it}. \quad (2)$$

For a variable z assigned a GMM-style role, Difference GMM uses level instruments whose admissible lags are recorded explicitly by the role- or variable-specific set $\mathcal{L}_z^D$:

$$\mathbb{E}[z_{i,t-s} \Delta \varepsilon_{it}] = 0, \quad s \in \mathcal{L}_z^D. \quad (3)$$

For the lagged dependent variable, the conventional lower bound is two. System GMM stacks Equation 2 with the level equation and, under the additional initial-condition restrictions of Blundell and Bond (1998), admits level-equation moments of the form

$$\mathbb{E}[\Delta z_{i,t-1} (\alpha_i + \varepsilon_{it})] = 0. \quad (4)$$

Thus the implementation constructs Z_i from declared roles and lag windows rather than silently selecting instruments. With $\boldsymbol{\theta} = (\rho, \boldsymbol{\beta}^\top)^\top$, the one- or two-step estimator minimizes

$$\begin{aligned} \mathbf{g}_N(\boldsymbol{\theta}) &= \frac{1}{N} \sum_{i=1}^N Z_i^\top \mathbf{u}_i(\boldsymbol{\theta}), \\ \hat{\boldsymbol{\theta}} &= \arg \min_{\boldsymbol{\theta}} \mathbf{g}_N(\boldsymbol{\theta})^\top \widehat{W}_N \mathbf{g}_N(\boldsymbol{\theta}). \end{aligned} \quad (5)$$

For two-step estimation, $\widehat{W}_N$ is updated from first-step residual moments; the Windmeijer adjustment changes the reported finite-sample covariance, not the point estimate. For an overidentified model with q instruments and k estimated parameters, the robust Hansen statistic is

$$J_N = N \mathbf{g}_N(\hat{\boldsymbol{\theta}})^\top \widehat{W}_N \mathbf{g}_N(\hat{\boldsymbol{\theta}}) \stackrel{a}{\sim} \chi_{q-k}^2. \quad (6)$$

This reference distribution requires valid moment conditions and regularity conditions. Accordingly, Hansen non-rejection and AR(2) non-rejection are reported as diagnostics, not treated as proofs of identification.

4. Reproducible output workflow

All tables and figures in this manuscript are generated from machine-readable artifacts. **Universal Output Hub** 0.2.4 is the single manuscript table exporter. The numerical chain is raw or controlled data, executable model code, normalized CSV or release certificate, OutputHub LaTeX fragment or scripted figure, and finally this document. All stochastic procedures use the fixed seed 20260724. Debug output, row-level panel previews, API snapshots, smoke-test status tables, duplicate manual renderings, and stale deterministic target-lag results are retained outside the manuscript or excluded entirely.

5. Leakage-controlled N-CMAPSS application

The N-CMAPSS DS01 development data are used as a compact application of panel preparation and late-cycle validation. Remaining useful life decreases deterministically with cycle, so including lagged RUL—or the derived lagged risk proxy—creates a tautological prediction problem. The revised design removes those target lags. It retains a lagged-sensor OLS baseline and adds a nonlinear direct sensor model whose scaler, principal components, and tuning are fitted only on training windows (Table 1).

Let $\mathbf{s}_{it}$ denote the raw sensor vector and let $\mathcal{P}_{\mathcal{T}}$ denote scaling followed by principal-component analysis fitted only on training window $\mathcal{T}$. The nonlinear predictor used in the final holdout comparison is

$$\begin{aligned} \mathbf{q}_{it} &= \mathcal{P}_{\mathcal{T}}(\mathbf{s}_{it}), \\ \hat{r}_{it} &= f_{\hat{\eta}}(\mathbf{q}_{it}, \mathbf{q}_{i,t-1}, \Delta \mathbf{q}_{it}, \bar{\mathbf{q}}_{it}^{(3)}), \end{aligned} \quad (7)$$

where r_{it} is the risk proxy and $f_{\hat{\eta}}$ is the selected random forest. Equation 7 contains no cycle, lagged RUL, or lagged risk term and is a predictive mapping rather than a structural or causal equation.

Table 2 reports pooled and unit fixed-effects baselines. The lagged sensor-state coefficient is positive in both models (0.032 and 0.028) and is distinguishable from zero at the 5 percent

Design item	Value
Dataset	N-CMAPSS DS01 dev unit-cycle panel
Analysis target	Risk proxy derived from RUL
Observations after lagging	547
Units	6
Cycle range	2–100
OLS regressors	Lagged sensor summary and operating controls
Direct-model regressors	Train-fitted sensor PCs and dynamics
Target-lag rule	Lagged risk/RUL excluded
Validation design	Nested selection through cycle 70; test 71–100

Table 1: Leakage-controlled N-CMAPSS application design. Source: Author’s computation from the processed N-CMAPSS DS01 panel (Artifacts 01 and 05). Notes: The target is a risk proxy derived from RUL. Lagged risk and lagged RUL are excluded because they encode a deterministic countdown identity.

term	Pooled OLS	Unit FE
Constant	0.537*** (0.007)	0.495*** (0.009)
Lagged sensor state	0.032** (0.012)	0.028** (0.012)
Flow control (Fc)	0.024** (0.012)	-0.012 (0.013)
Health setting (hs)	-0.207*** (0.006)	-0.209*** (0.006)
N	547	547
R2	0.584	0.604
Adj. R2	0.582	0.598

Table 2: N-CMAPSS lagged-sensor baseline estimates. Source: Author’s computation using statsmodels with HC1 covariance (Artifacts 02 and 03). Notes: HC1 standard errors are in parentheses. Unit fixed effects are included in the second column but their individual coefficients are omitted from display.

level. The health-setting coefficient is negative and stable across specifications. By contrast, the flow-control coefficient is positive in pooled OLS but becomes small and statistically indistinguishable from zero after unit effects are introduced. In-sample fit rises only modestly, from 0.584 to 0.604, which is consistent with some unit heterogeneity but does not establish predictive generalization.

The late-cycle comparison withholds cycles 71–100 (Table 3). The pooled lagged-sensor baseline has RMSE 0.332 and out-of-sample R^2 -15.783; its squared error exceeds that of a constant benchmark equal to the observed holdout mean. The direct random forest instead attains RMSE 0.071, MAE 0.053, SMAPE 0.062, and R^2 0.232. Candidate selection uses three expanding windows entirely within cycles 1–70, and the scaler and PCA are refitted in every

Model	Split	N eval	RMSE	MAE	SMAPE	R2
Pooled dyn. OLS	Cycles 1–70 / 71–100	133	0.332	0.323	0.447	-15.783
Direct sensor RF	Cycles 1–70 / 71–100	133	0.071	0.053	0.062	0.232

Table 3: Late-cycle holdout performance. Source: Author’s computation from the nested cycle-ordered validation (Artifacts 04, 06, and 07). Notes: The forest is selected only within cycles 1–70 using expanding windows. The final cycles 71–100 are evaluated once; neither model uses lagged risk, lagged RUL, or cycle.

training window before the final outer evaluation. The improvement is predictive evidence for this split, not a causal claim or proof that late-life distribution shift is absent.

6. Controlled dynamic-GMM results

The controlled panel contains 120 units observed over ten periods. The data-generating coefficients are 0.55 for persistence, 0.35 for the predetermined regressor, and 0.25 for the exogenous regressor. Both estimators use collapsed instruments and two-step covariance calculations. Instrument counts (five and eight) remain well below the number of groups.

The exact controlled data-generating process is

$$y_{it} = 0.55y_{i,t-1} + 0.35x_{it}^{\text{pred}} + 0.25x_{it}^{\text{exog}} + \alpha_i + \tau_t + \varepsilon_{it}. \quad (8)$$

The predetermined regressor depends on the previous-period disturbance but not the current disturbance; the exogenous regressor is generated independently.

Difference GMM estimates the exogenous effect at 0.251, close to its generating value, while its persistence estimate is 0.711 with a comparatively large standard error. System GMM estimates the exogenous effect at 0.242 but produces a higher persistence estimate (0.929) and a weak predetermined-regressor estimate (0.056). These differences show why estimator choice cannot be reduced to in-sample precision. In this controlled realization, System GMM is not uniformly closer to the generating parameters.

Both models show the expected first-order serial correlation after differencing and do not reject AR(2) at conventional levels. Hansen p-values are 0.503 and 0.794. These diagnostics are compatible with the maintained instrument design, but non-rejection is not proof of instrument exogeneity and a high Hansen p-value can be uninformative when instruments proliferate (Roodman 2009).

7. Cross-software validation

The formal System GMM evidence in release 0.5.14 comprises six collapsed two-step specifications. Across them, native and **Stata** results have identical observation, group, instrument, and over-identification counts. Maximum absolute coefficient differences are below the registered 10^{-6} tolerances, and relative standard-error differences remain within each specification’s registered tolerance (Table 5). The unbalanced and variable-missing fixtures also match exact panel-time estimation-sample keys.

term	Difference GMM	System GMM
Lagged dependent variable	0.711*** (0.265)	0.929*** (0.042)
Predetermined regressor	0.266 (0.223)	0.056 (0.138)
Exogenous regressor	0.251*** (0.045)	0.242*** (0.020)
Constant		0.129*** (0.035)
N	840	960
Groups	120	120
Instruments	5	8
I/N	0.042	0.067
Instrument health	acceptable	acceptable
AR(1) p	0.000	0.000
AR(2) p	0.133	0.199
Hansen p	0.503	0.794
Sargan p	0.864	0.812

Table 4: Controlled Difference GMM and System GMM results. Source: Author’s computation with *systemgmmkit* 1.0.0 (Artifact 22). Notes: Two-step collapsed specifications. Standard errors are in parentheses. AR and over-identification entries are p-values; diagnostic non-rejection is not proof of validity.

Specification	N	Groups	Instr.	Max coef. diff	Max rel. SE diff	Keys	Status
Baseline controls	1248	96	8	2.31e-14	5.78e-07	Not required	PASS
No controls	1248	96	7	8.88e-16	5.46e-07	Not required	PASS
Three-way controls	1248	96	16	3.66e-15	2.63e-06	Not required	PASS
Decomposition controls	1248	96	12	1.89e-14	1.57e-07	Not required	PASS
Unbalanced panel	1174	96	8	1.04e-14	1.54e-06	Exact	PASS
Variable missingness	1216	96	8	2.29e-14	2.16e-06	Exact	PASS

Table 5: System GMM parameter and count parity with Stata *xtabond2*. Source: Author’s computation from the signed v0.5.14 release certificate (Artifact 24). Notes: All six release-certified specifications pass their registered tolerances. Exact sample-key matching is required for the unbalanced and variable-missing designs; Not required means that gate was not registered for the four balanced fixtures.

Table 6 separates numerical agreement from empirical adequacy. Baseline and unbalanced designs do not reject the over-identifying restrictions at 5 percent, whereas several deliberately maintained fixtures do. Those rejections are preserved because a parity test must reproduce the comparator’s result even when the specification would not be selected for substantive use. AR(2) is not rejected in any of the six registered specifications.

Specification	Hansen p	Sargan p	AR(1) p	AR(2) p	Over-id (5%)	Status
Baseline controls	0.160	0.088	0.080	0.868	Do not reject	PASS
No controls	0.024	0.006	0.136	0.920	Reject	PASS
Three-way controls	0.021	0.000	0.000	0.990	Reject	PASS
Decomposition controls	0.006	0.000	0.064	0.152	Reject	PASS
Unbalanced panel	0.232	0.297	0.107	0.873	Do not reject	PASS
Variable missingness	0.012	0.001	0.033	0.111	Reject	PASS

Table 6: System GMM diagnostic parity with Stata `xtabond2`. Source: Author’s computation from the signed v0.5.14 release certificate (Artifact 24). Notes: PASS denotes cross-software numerical agreement. The over-identification column reports the Stata-side 5 percent decision and is not a model endorsement.

Estimator	Comparators	Max coef. diff	Max SE diff
OLS	R <code>lm</code> , Stata, <code>statsmodels</code>	2.16e-08	4.01e-10
Pooled OLS	R <code>plm</code> , Stata, <code>linearmodels</code>	2.16e-08	4.01e-10
Fixed Effects	R <code>plm</code> , Stata, <code>linearmodels</code>	1.60e-08	1.14e-09
Random Effects	R <code>plm</code> , Stata, <code>linearmodels</code>	2.16e-08	6.58e-04
2SLS	R <code>AER::ivreg</code> , Stata, <code>linearmodels</code>	6.35e-08	3.80e-04

Table 7: Aligned static-estimator reference checks. Source: Author’s computation from the v0.5.14 cross-software checks (Artifact 27). Notes: Specifications and samples are aligned. Random-effects and 2SLS claims remain at the coefficient level because covariance conventions differ across packages.

Static reference checks tell a similarly qualified story. OLS, pooled OLS, and fixed-effects results agree numerically with aligned Python/R references and closely with Stata exports. Random-effects and 2SLS coefficients also agree, but standard errors vary modestly under package-specific covariance conventions. Table 7 therefore labels those rows as coefficient agreement rather than overstating full covariance parity.

8. Post-estimation, ML-style workflows, and automatic GMM search

The ML namespace is an orchestration layer around fitted econometric results. It supplies panel-aware splitting, prediction metrics, model comparison, forecasting, and expanding-window backtesting. These tools help distinguish estimation fit from predictive performance, but predictive accuracy does not create causal identification (Roberts, Bahn, Ciuti, Boyce, Elith, Guillera-Arroita, Hauenstein, Lahoz-Monfort, Schröder, Thuiller, Warton, Wintle, Hartig, and Dormann 2017; Bergmeir, Hyndman, and Koo 2018; Cerqueira, Torgo, and Mozetič 2020).

The automatic GMM search layer explores user-declared Difference/System GMM candidates over lag windows, transformation, estimation step, collapse choice, and related options. Candidate ranking occurs only after diagnostic gates for convergence, AR(2), Hansen behavior, and instrument proliferation. This design reduces mechanical specification shopping, but

Candidate	Status	RMSE	R2	Hansen p	AR(2) p	Instr.
System, 2:2	Selected	0.602	0.923	0.669	0.287	6
System, 2:3	Admissible	0.606	0.922	0.929	0.286	8
Difference, 2:2	Rejected	1.877	0.251	Not reported	0.139	3
Difference, 2:3	Admissible	1.382	0.594	0.897	0.151	5

Table 8: Diagnostic-first automatic GMM search on the controlled panel. Source: Author’s computation from the executed diagnostic-first search. Notes: Cycle-ordered holdout covers the final two periods. The strict gate rejects the Difference GMM 2:2 candidate because a Hansen test is not reported; the value is not imputed. Selection ranks only admissible candidates and does not establish causal identification.

it cannot supply substantive exclusion restrictions or decide which moment conditions are credible. The final specification remains a research decision.

For the executed strict-policy search, the diagnostic filter and predictive selection can be summarized as

$$\mathcal{A} = \left\{ m \in \mathcal{M} : \text{fit}_m = \text{ok}, 0.05 < p_{H,m} < 0.99, p_{\text{AR}(2),m} > 0.05, j_m \leq n_m \right\}, \quad (9)$$

$$\hat{m} = \arg \min_{m \in \mathcal{A}} \text{RMSE}_m.$$

Here j_m and n_m are the instrument and group counts. Missing required diagnostics exclude a candidate under the strict policy. These thresholds are configurable API defaults and should not be interpreted as universal statistical decision rules.

The executed four-candidate search selects the System GMM 2:2 lag window. Its cycle-ordered holdout RMSE is 0.602 and its out-of-sample R^2 is 0.923. The System GMM 2:3 candidate is also admissible and performs similarly, while the admissible Difference GMM 2:3 candidate has a markedly larger RMSE of 1.382. The Difference GMM 2:2 candidate is rejected by the strict gate because its exactly identified design does not provide a Hansen test. This is a policy choice for comparable diagnostic screening, not evidence that every exactly identified model is intrinsically invalid.

A repository-only comparison artifact places this selected candidate beside external Python predictive baselines on the identical final-two-period holdout. The **statsmodels** OLS baseline records RMSE 0.597 and R^2 0.924; the **scikit-learn** random forest records RMSE 0.655 and R^2 0.909. The artifact is not reproduced as a manuscript table. It checks temporal-split and predictive-metric behavior only, not estimator parity, Hansen/AR agreement, or causal identification.

9. Limitations

The N-CMAPSS illustration has only six units after processing and is too small to serve as a formal dynamic-GMM application. It is retained to demonstrate leakage control and honest temporal validation. The controlled GMM exercise is synthetic and establishes workflow behavior, not external validity. Cross-software certificates are specification-specific, and their tolerances, samples, comparator versions, and provenance must accompany any parity claim. Finally, no diagnostic or automated search rule can replace an argument for identification.

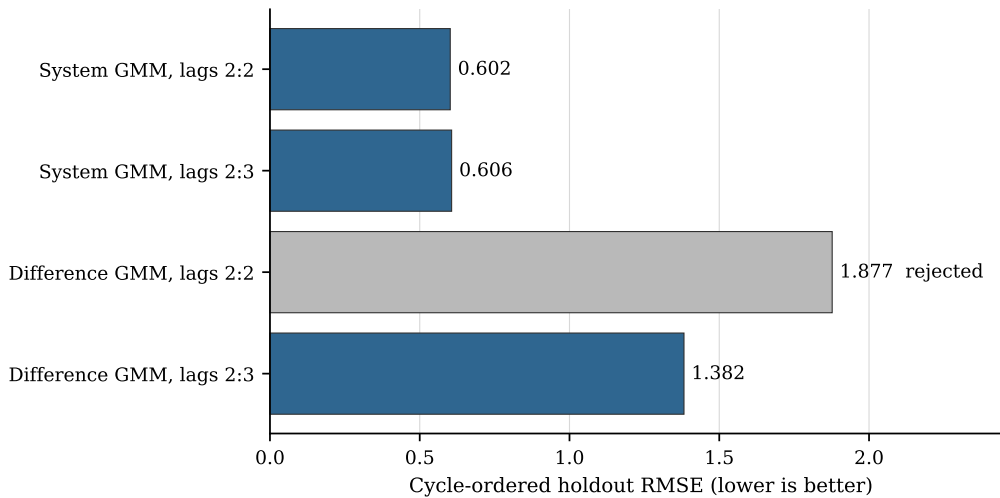


Figure 1: Cycle-ordered holdout error for the four automatic GMM-search candidates. Source: Author’s computation from the executed search in Table 8. The gray candidate failed the strict diagnostic-availability gate; lower RMSE is better, and predictive ranking does not establish causal identification.

Layer	APIs/workflow	Runtime	Boundary
Post-estimation	predict, confint, margins	API verified	Estimator-based inference
Panel validation	split, metrics	Smoke passed	Panel-time order
Forecasting	forecast, backtest, compare	API verified	No causal claim

Table 9: Post-estimation and ML-style workflow coverage. Source: Author’s runtime audit of the installed systemgmmkit 1.0.0 API. Notes: This is workflow coverage, not estimator parity. Prediction and validation utilities operate around fitted econometric results and do not create causal identification.

10. Availability and reproducibility

The source code and release evidence are available from the project repository under the MIT license. The separate JOSS paper remains in `paper/paper.md`. This JSS manuscript and its OutputHub-generated tables live only in `paper_jss/`. Running `run_all.py` in open mode recreates data products, model outputs, comparisons, all nine tables, the figure, the manuscript PDF, logs, checksums, and the reproducibility manifest. The build refuses to run unless the installed package is exactly version 1.0.0 and the table exporter is exactly **Universal Output Hub** 0.2.4. The six-specification parity tables are read from the signed v0.5.14 Git tag rather than from mutable development artifacts. The portable manual Stata script and the repository-only external Python comparison are retained under `replication/scripts/` with machine-readable outputs for future independent review.

AI assistance disclosure

Generative AI tools assisted with software workflow planning, code review, table-pipeline

construction, and drafting/editing of technical prose. The author is responsible for verifying every result, revising the text, completing the literature-led discussion, and ensuring that the submitted manuscript satisfies journal policy.

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